

Catherine Arnett

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Employment

EleutherAI, *remote*

NLP Researcher

2025 -

PleIAs, *Paris, France (remote)*

Research Scientist

2024-2025

Research Intern

2024

Education

PhD, Linguistics, UC San Diego

2019-2025

With a specialization in Computational Social Science

Dissertation: “[A Linguistic Approach to Crosslingual and Multilingual NLP](#)”

Committee: Farrell Ackerman (chair), Benjamin Bergen, Leon Bergen, Victor Ferreira

MA (hons) Chinese and Linguistics, First Class Honours, University of Edinburgh

2014-2018

Including one year exchange at Zhejiang University, Hangzhou, P.R. China

Publications

High-Impact Conference Proceedings

1. Tyler A. Chang*, **Catherine Arnett***, and 300+ others (2026). [Global PIQA: Evaluating Physical Commonsense Reasoning Across 100+ Languages and Cultures](#). *Equal contribution. Fortieth Annual Conference on Neural Information Processing Systems (NeurIPS), Evaluations and Benchmarks Track. Sydney, Australia.
2. Guijin Son, Seungone Kim, **Catherine Arnett**, and 63 others. (under review) [Soohak: An Mathematician-curated benchmark for Assessing Research-level Math Capabilities of LLMs](#). Fortieth Annual Conference on Neural Information Processing Systems (NeurIPS), Evaluations and Benchmarks Track. Sydney, Australia.
3. Xiulin Yang, Ethan Gotlieb Wilcox, **Catherine Arnett** (2026). [Apples to Apples? Towards Comparable Crosslingual Language Model Evaluation](#). Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing. Budapest, Hungary.
4. Suchir Salhan, **Catherine Arnett**, James A. Michaelov, and Paula Buttery (2026). [Beetle: Structured Exposure Pre-training in Bilingual Language Models for Modelling L2 Language Processing](#). Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing. Budapest, Hungary. **Selected for Oral.**
5. Ej Zhou, Suchir Salhan, **Catherine Arnett**, and Anna Korhonen (2026). [Cross-Lingual Alignment Without Joint Training: Do Monolingual Language Models Converge on Universal Representations?](#) Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing. Budapest, Hungary.
6. Antonio Lopardo, Luis Frentzen Salim, Shikhar Srivastava, Kevin Jiang, Mridul Sharma, **Catherine Arnett** (2026). No Single Tokenizer Feature Reliably Predicts Downstream Language Model Performance. Findings of the Association for Computational Linguistics: EMNLP 2026. Budapest, Hungary.
7. Shivam Singh, Aditya Yadavalli, **Catherine Arnett**, and Alex Warstadt (2026). [BuzzASR: A Swarm of 100+ Monolingual Speech Recognition Models](#). Findings of the Association for Computational Linguistics: EMNLP 2026. Budapest, Hungary.
8. Stella Biderman, Mohammad Aflah Khan, Niloofar Mireshghallah, **Catherine Arnett**, Fazl Barez, Naomi Saphra (2026). [Position: Don't Just "Fix it in Post": A Science of AI Must Study Learning Dynamics](#). Forty-Third International Conference on Machine Learning (ICML), Position Paper Track. Seoul, South Korea. **Selected for Oral (top 1% of submissions).**

9. Pedro Ortiz Suarez*, Laurie Burchell*, **Catherine Arnett***, and 94 other (2026). [CommonLID: Re-evaluating State-of-the-Art Language Identification Performance on Web Data](#). Proceedings of the Association for Computational Linguistics. San Diego, USA. *Equal contribution.
10. Antonio Lopardo, Avyukth Harish, **Catherine Arnett**, Akshat Gupta (2026). [Weight Tying Biases Token Embeddings Towards the Output Space](#). Findings of the Association for Computational Linguistics. San Diego, USA.
11. Tyler A. Chang, **Catherine Arnett**, Zhuowen Tu, Benjamin K. Bergen (2026). [Goldfish: Monolingual Language Models for 350 Languages](#). Language Resources and Evaluation Conference (LREC). Palma de Mallorca, Spain. **Selected for Oral**.
12. Pierre-Carl Langlais, Pavel Chizhov, **Catherine Arnett**, Carlos Rosas Hinostroza, Mattia Nee, Eliot Krzystof Jones, Irène Girard, David Mach, Anastasia Stasenko, and Ivan P. Yamshchikov (2026). [Common Corpus: The Largest Collection of Ethical Data for LLM Pre-Training](#). The International Conference on Learning Representations (ICLR). Rio de Janeiro, Brazil. **Selected for Oral (top 1% of submissions)**.
13. **Catherine Arnett**, Tyler A. Chang, Stella Biderman, Benjamin K. Bergen (2025). [Explaining and Mitigating Crosslingual Tokenizer Inequities](#). The Thirty-Ninth Annual Conference on Neural Information Processing Systems (NeurIPS). San Diego, CA, USA.
14. **Catherine Arnett**, Tyler A. Chang, James A. Michaelov, and Benjamin K. Bergen (2025). [On the Acquisition of Shared Grammatical Representations in Bilingual Language Models](#). The 63rd Annual Meeting of the Association for Computational Linguistics (ACL). Vienna, Austria.
15. **Catherine Arnett** and Benjamin K. Bergen (2025). [Why do language models perform worse for morphologically complex languages?](#) The 31st International Conference on Computational Linguistics (COLING). Abu Dhabi, UAE and online. **Best Paper Award (top 0.25% of accepted papers)**.
16. Pavel Chizhov*, **Catherine Arnett***, Elizaveta Korotkova, Ivan P. Yamshchikov (2024). [BPE Gets Picky: Efficient Vocabulary Refinement During Tokenizer Training](#). Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP). Miami, FL, USA. *equal contribution.
17. Tyler A. Chang, **Catherine Arnett**, Zhuowen Tu, Benjamin K. Bergen (2024). [When Is Multilinguality a Curse? Language Modeling for 250 High- and Low-Resource Languages](#). Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP). Miami, FL, USA. **Outstanding Paper Award (top 2.5% of accepted papers)**.
18. James Michaelov, **Catherine Arnett**, Benjamin K. Bergen (2024). [Revenge of the Fallen? Recurrent Models Match Transformers at Predicting Human Language Comprehension Metrics](#). The First Conference on Language Modeling (COLM). Philadelphia, PA, USA.
19. James A. Michaelov*, **Catherine Arnett***, Tyler A. Chang, Benjamin K. Bergen (2023). [Structural priming demonstrates abstract grammatical representations in multilingual language models](#). Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP). Singapore. *equal contribution
20. **Catherine Arnett** (2019). [Pathways of Change in Romance Motion Events: A Corpus-based Comparison](#). *Proceedings of the Thirtieth Western Conference on Linguistics (WeCOL)*. Vol 23. Fresno, CA, USA.

Archival Workshop Proceedings

1. Laura Barbenel, Lily Goulder, Aoife O'Driscoll, Suchir Salhan, **Catherine Arnett**, Andrew Caines, Paula Buttery (2026). [L1 Influence in L2 Language Models: A Human-centric Approach](#). [Workshop on Computational Developmental Linguistics](#). Co-located with ACL. San Diego, USA.
2. **Catherine Arnett**, Marisa Hudspeth, and Brendan O'Connor (2025). [Evaluating Morphological Alignment of Tokenizers in 70 Languages](#). Tokenizer Workshop at ICML. Vancouver, Canada.
3. Sander Land and **Catherine Arnett** (2025). [BPE Stays on SCRIPT: Structured Encoding for Robust Multilingual Pretokenization](#). Tokenizer Workshop at ICML. Vancouver, Canada. **Best Paper Award**.
4. **Catherine Arnett***, Pamela D. Rivière*, Tyler A. Chang, and Sean Trott (2024). [Different Tokenization Schemes Lead to Comparable Performance in Spanish Number Agreement](#). Special Interest Group on Computational Morphology and Phonology (SIGMORPHON) at NAACL. Mexico City, Mexico. *equal contribution

5. **Catherine Arnett***, Tyler A. Chang*, Benjamin K. Bergen (2024). [A Bit of a Problem: Measurement Disparities in Dataset Sizes Across Languages](#). 3rd Annual Meeting of the Special Interest Group on Under-resourced Languages (SIGUL), at LREC-COLING. Torino, Italy. *equal contribution

Refereed Journal Articles

1. Jesse Quinn, Matthew Goldrick, **Catherine Arnett**, Victor S. Ferreira, Tamar H. Gollan (2024). Syntax Drives Default Language Selection in Bilingual Connected Speech Production. *Journal of Experimental Psychology: Learning, Memory, and Cognition*.

Conference Presentations

1. Suchir Salhan, **Catherine Arnett**, James A. Michaelov, and Paula Buttery (2026). Structured Exposure Pretraining in Bilingual Language Models for Modelling L2 Reading. Computational Psycholinguistics Meeting. Utrecht University. **Selected for Talk**.
2. Suchir Salhan, **Catherine Arnett**, James A. Michaelov, and Paula Buttery (2026). Structured Exposure Pretraining in Bilingual Language Models for Modelling L2 Reading. Crosslinguistic Perspectives on Processing and Learning (X-PPL). Bayonne, France.
3. Eugene Jang, **Catherine Arnett**, and Terra Blevins (2026). What Makes an Encoding Good for Language Modeling? Tokenizer Workshop at the Conference on Language Models (COLM). San Francisco, California.
4. **Catherine Arnett** and Benjamin K. Bergen (2026). Structural Priming Effects in Language Models are Less Human-like in Languages Other Than English. Human Sentence Processing Conference (HSP). MIT. **Selected for Talk (10% of submissions)**.
5. Suchir Salhan, Ej Zhou, Laura Barbenel, Aoife O'Driscoll, Lily Goulder, Lucas Resck, **Catherine Arnett**, Paula Buttery (2026). Glints of Gold or Troubling Waters? Can a School of Merged Monolingual Goldfish Models Swim in Bilingual Seas?. First Workshop on Multilingual Multicultural Evaluation at EACL. Rabat, Morocco.
6. Sander Land and **Catherine Arnett** (2025). [BPE Stays on SCRIPT: Structured Encoding for Robust Multilingual Pretokenization](#). EurIPS.
7. James A. Michaelov and **Catherine Arnett** (2025). [Benchmark Disaggregation Improves Interpretability of Training Dynamics](#). Interpreting Cognition in Deep Learning Models (CogInterp) Workshop at NeurIPS. San Diego, CA, USA.
8. Jennifer Meng Lu, **Catherine Arnett**, Ruochen Zhang. (2025) Mechanisms of In-Context Syntactic Generalization in Language Models. Blackbox NLP @ EMNLP. Suzhou, China.
9. Jennifer Meng Lu, **Catherine Arnett**, Ruochen Zhang. (2025) Mechanisms of In-Context Syntactic Generalization in Language Models. New England Mechanistic Interpretability Workshop (NEMI). Northeastern University.
10. **Catherine Arnett** (2025). [Toxic Commons: Curating Open-Source Pre-Training Data](#). *The First Conference of the International Association for Safe & Ethical AI*. Official event of the AI Action Summit. Paris, France.
11. **Catherine Arnett**, Tyler A. Chang, James A. Michaelov, and Benjamin K. Bergen (2023). [Crosslingual Structural Priming and the Pre-Training Dynamics of Bilingual Language Models](#). *The 3rd Workshop on Multilingual Representation Learning co-located with EMNLP 2023*. Singapore.
12. **Catherine Arnett** & Maho Takahashi (2022). Creating a Baseline to Evaluate Correlations Between Language and Environment. *Machine Learning and the Evolution of Language*. Kanagawa/online. [\[abstract\]](#) [\[poster pdf\]](#)
13. **Catherine Arnett** & Eva Wittenberg (2020). Multiple Meanings of Doubling Up: Mandarin Verbal Reduplication. *The 26th Architectures And Mechanisms for Language Processing Conference (AMLaP)*. Potsdam/online. [\[poster pdf\]](#)
14. **Catherine Arnett** & Eva Wittenberg (2020). Conceptual Effects of Verbal Reduplication in Mandarin Chinese. *North American Conference on Chinese Linguistics 32*. Storrs, CT/online.
15. **Catherine Arnett** & Eva Wittenberg (2019). Conceptual Effects of Verbal Reduplication in Mandarin Chinese. California Meeting on Psycholinguistics. Santa Cruz, California. [\[poster pdf\]](#)
16. **Catherine Arnett** (2018). Diachronic study of the typology of motion verbs in the Romance languages. Undergraduate Linguistics Association of Britain Conference, University of Edinburgh.

Manuscripts

1. Eugene Jang, **Catherine Arnett**, and Terra Blevins (in prep). Alternative Encodings for Byte-Level Modeling
2. **Catherine Arnett**, Jiani Zhong, Nora Tseng, Suchir Salhan, Alex Warstadt, Benjamin K. Bergen (in prep). How to Build a BLiMP: Best Practices for Creating Linguistic Minimal Pair Benchmarks.
3. Marco Cognition, Christopher Akiki, Pawan Sasanka Ammanamanchi, **Catherine Arnett**, ..., and Yuval Pinter (under review). [Tokenization: A Survey for Modern NLP](#).
4. Bobby Cheng, Adam Gaber, Zhengyuan Liu, **Catherine Arnett**, Omer Goldman, Cheston Tan, Leshem Choshen (under review). [Skill Issue: Are Skills Language-Invariant in LLMs?](#).
5. Xiulin Yang, **Catherine Arnett**, and Ethan Gotlieb Wilcox (under review). Autoregressive Transformer Models Are Systemically Biased Toward English-like Languages.
6. James A. Michaelov, **Catherine Arnett**, Benjamin K. Bergen (under review). Language Model Shape Shapes Behavior.
7. James A. Michaelov, **Catherine Arnett**, Pamela D. Riviere, Tyler A. Chang, Samuel M. Taylor, Cameron Robert Jones, Sean Trott, Roger P. Levy, Benjamin K. Bergen, Micah Altman (under review). [How Open Must Language Models be for Reliable Scientific Research?](#)

Invited Talks

Multilingual Base Model Evaluation

Collin Raffel's lab, University of Toronto

September 2026

Explaining and Mitigating Crosslingual Tokenizer Inequities

Keynote, Tokenization Seminar

May 2026

Department of Information and Computing Sciences, Utrecht University

Linguistics in the Age of LLMs [\[slides\]](#)

Guest Lecture, *Introduction to Linguistics*, Harvard University

July 2025

Glitches in the Embedding Matrix [\[slides\]](#)[\[video\]](#)

EleutherAI

June 2025

When is Multilinguality a Curse? Language Models for 350 Languages [\[slides\]](#)

Cambridge NLIP Seminar Series, Cambridge University

June 2025

Best Practices for Open Multilingual LLM Evaluation

PyTorch Day, GOSIM AI. Paris, France.

May 2025

Why do language models perform worse for morphologically complex languages?

NLP Seminar, University of Massachusetts Amherst

September 2025

tinlab, Boston University

March 2025

Characterizing Shared Multilingual Representations with Structural Priming

Brown NLP, Brown University

March 2025

Media and Outreach

NLP Blog Posts

[There is no such thing as a tokenizer-free lunch](#)

[An Analysis of Multilingual Models on Hugging Face](#)

[Best Practices for Open Multilingual LLM Evaluation](#)

[Small is Beautiful](#)

[wHy DoNt YoU jUsT uSe ThE lLaMa ToKeNiZeR??](#)

Author

[Cognitive Foundations, an open Textbook](#)

Update and rewrite portions of the *Language* chapter using bookdown in R.

Teaching Experience

Graduate Teaching Consultant for International Instructors , UC San Diego	2022-2024
Instruct incoming international graduate students in teaching skills and subject-specific English	
Instructor of Record , UC San Diego	Summer 2023
LIGN 170: Psycholinguistics	
Course Website	
Teaching Assistant , UC San Diego	2019-2022
LIGN 101: Introduction to Language, Fall 2020-Spring 2022	
LIDS 19: Independent Language Study, Winter 2020-Spring 2020	
LIGN 170: Psycholinguistics, Fall 2019	

Selected Scholarships, Fellowships, and Grants

Language Equity & Inclusion through AI (EquILL-AI), University of Cambridge, £5,750	2026
Yankelovich Graduate Research Funding, UC San Diego, \$3,500	2024
Linguistics Department Anti-Racist Pedagogy Micro-Fellowship, UC San Diego Linguistics, \$500	2023
Summer Graduate Teaching Scholar, UC San Diego, \$1,000	2023
LaVerne Noyes Foundation Endowed Fellowship, UC San Diego, \$6,000	2022
Academic Senate Grant, UC San Diego, \$9,000	2022
Friends of the International Center Fellowship, UC San Diego, \$2000	2020
Eric Liddell China Saltire Scholarship, University of Edinburgh, £5,000	2016
St. Andrews Society of North Carolina Scholarship, \$37,000	2014-2018

Awards and Certifications

Outstanding Reviewer, CoNLL @ ACL	2026
Best Paper Award, Tokenization Workshop at ICML	2025
Best Paper Award, COLING	2025
Outstanding Paper Award, EMNLP	2024
Teaching Assistant Excellence Award, UCSD Linguistics	2022-2023
Edinburgh Award, Peer Learning and Support	2018

Skills and Languages

Languages

English - Native
Mandarin - Proficient
Spanish - Proficient

Academic Service

Organizing

Shared Task Organizer, Multilingual Representation Learning Workshop, EMNLP	2025-2026
Shared Task Organizer, Workshop on Data Quality in Multilingual Pre-Training Data, COLM	2025
Local organizer, EvalEval (unofficial workshop at NeurIPS)	2025

Reviewing

Area Chair

LREC	2026
ACL ARR	Feb, May 2025

Program Chair

Multilingual Representation Learning Workshop (MRL), EMNLP	2025-2026
Workshop on Multilingual Data Quality Signals (WMDQS), COLM	2025

Journal Reviewer

Computational Linguistics	2026
Natural Language Processing Journal	2026
Language Resources and Evaluation	2025
Journal of Cognitive Science	2025

Conference Reviewer

ACL ARR	2025-2026
NeurIPS	2025-2026
COLM	2025-2026
Cognitive Science Society	2026
LREC	2026
COLING	2025
Undergraduate Linguistics Association of Britain	2020

Workshop Reviewer

Web-as-Corpus Workshop, EMNLP	2026
Conference on Computational Natural Language Learning (CoNLL), ACL	2026
Workshop on Evaluating Evaluations (EvalEval), ACL	2026
CogInterp: Interpreting Cognition in Deep Learning Models, NeurIPS	2025
Workshop on Multilingual Data Quality Signals (WMDQS), COLM	2025
Workshop on Multilingual and Equitable Language Technologies (MELT), COLM	2025
Student Research Workshop, ACL	2025
Tokenizer Workshop, ICML and COLM	2025-2026
Multilingual Representation Learning Workshop (MRL), EMNLP	2024-2026

Ethics Reviewer

ACL ARR	2024-2026
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Advising**PhD Committee Member**

Marisa Hudspeth, University of Massachusetts Amherst	2025-
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Undergraduate Honors Theses Committee

Yuhan Fu, Cognitive Science B.S. Honors Thesis, UC San Diego	2024
“Is There Evidence for Embodied Simulation During Language Translation?”	

Professional Memberships

ELRA	2026-
Association for Computational Linguistics	2023-